

BENCHMARK: MARGINAL

Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460358.7412 +/- 0.0047 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.164 +/- 0.03                  |
| Transit depth (Rp/Rs)^2                           | 2.69 +/- 1.0 [%]                |
| Orbital Inclination (inc)                         | 81.88 +/- 2.12                  |
| Airmass coefficient 1 (a1)                        | 1.14 +/- 0.026                  |
| Airmass coefficient 2 (a2)                        | -0.091 +/- 0.027                |
| Scatter in the residuals of the lightcurve fit is | 2.3 %                           |
| Best Comparison Star                              | #1 - [334, 182]                 |
| Optimal Aperture                                  | 2.68                            |
| Optimal Annulus                                   | 12.31                           |
| Transit Duration (day)                            | 0.062 +/- 0.024                 |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.164 ± 0.03 vs 0.1368 (deviation 0.91σ) |
| Duration fit vs published | 0.062 d vs 0.0758 d (deviation 0.58σ)    |
| Mid-time O–C              | 13.6 min                                 |
| Depth SNR                 | 5.5                                      |
| Residual scatter          | 2.3 %                                    |

Light curve

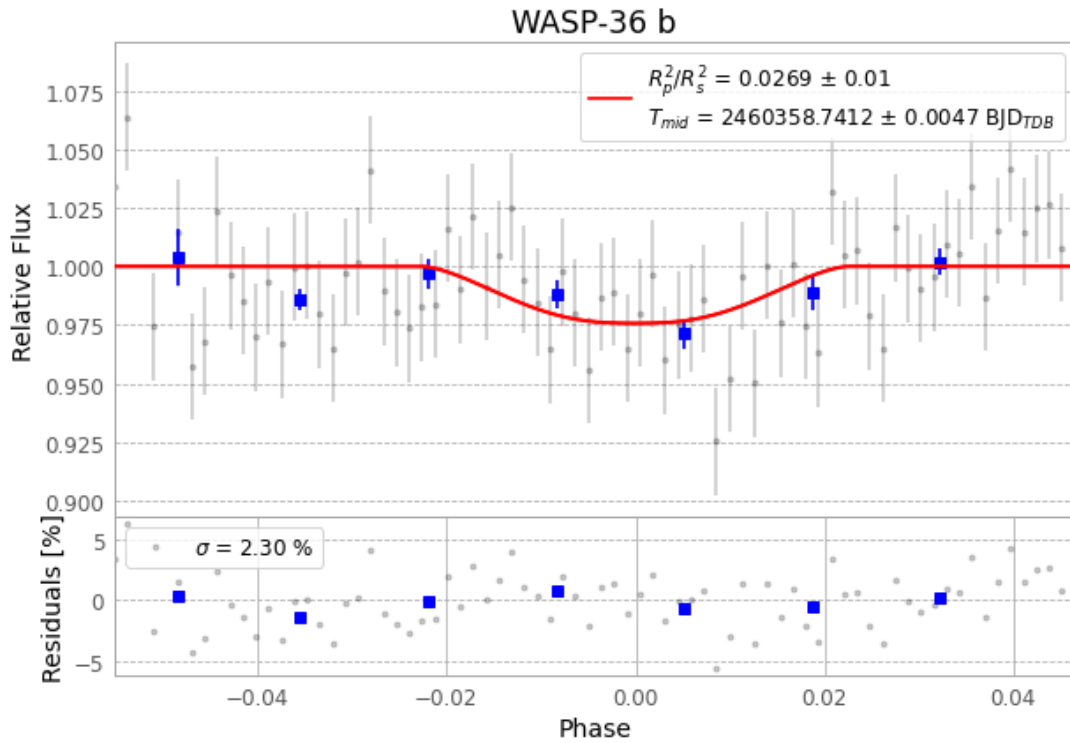


Figure 1 — FinalLightCurve\_WASP-36 b\_2024-02-17.png (SHA-256 ffff10c7132343e8...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [419, 228], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 558c5222ed94000f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ead60b9

## Data provenance

77 FITS frames (51 MB), WASP-36240218033315.FITS → WASP-36240218072115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-240218.log (2dc8bfddc805...), FinalLightCurve\_WASP-36 b\_2024-02-17.png (ffff10c71323...), FinalLightCurve\_WASP-36 b\_2024-02-17.csv (b38b0b5d71c3...)

## Compute resources

Wall time 225.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 09:48Z.